Adarsh Singh

Software Engineer | AI & Security

+91 7439473588 | **Kolkata, India** | **adarsh042021@outlook.com** | **LinkedIn** | **GitHub**

Education

Lovely Professional University, Jalandhar, Punjab — **August 2022 – July 2026**
Bachelor of Technology — Computer Science and Engineering (Minor: Cyber Security)
CGPA: 7.27

Experience

Titan Company Limited — *12 Jan 2026 – 10 Aug 2026*
Intern | InfoSec Department — *Onsite, Bengaluru*

WAF Automation · Node.js, Playwright, JavaScript

- Eliminated manual UI navigation across **170+ WAF-protected domains** by building a headless Node.js/Playwright RPA tool with SSO-aware Edge profiles for automated report downloading.
- Engineered virtualized dropdown discovery, session keep-alive mechanisms, and automated CSV audit logging organized by month for full operational traceability.

VAPT Assessment Portal · React, TypeScript, Node.js, Express, Prisma, MySQL

- Reduced report generation time from **7 hours to <1 hour** by architecting an enterprise portal replacing manual Excel workflows across **5 assessment types** with a 14-table data model.
- Integrated an enterprise LLM gateway (JWT auto-refresh) to auto-populate CVSS/CWSS scores, CWE identifiers, and remediation narratives directly from analyst PoCs.
- Built a compliance-grade vulnerability lifecycle featuring append-only audit tables, SLA breach tracking, executive risk heatmaps, and asynchronous PDF generation via BullMQ and Puppeteer.

Large-Scale Data Diffing Tool · React, TypeScript, Node.js, Express, TanStack Virtual, Zustand, Microsoft Graph API

- Built a full-stack dashboard ingesting two Excel reports (**up to 100MB each**), running a custom diffing algorithm to classify records as new, resolved, or persistent across scans.
- Rendered **300,000+ rows** smoothly via TanStack Virtual and integrated Microsoft Graph API to auto-notify relevant owners by email with delivery tracking logs.

Projects

Adarshclaw — Secure Human-in-the-loop AI Assistant — *Oct 2026*
TypeScript, Node.js, Bun, Vercel AI SDK, Docker

- Engineered a secure AI coding assistant supporting autonomous task execution, codebase analysis, and isolated web research featuring a **human-in-the-loop approval workflow** and **Telegram remote control**.
- Architected defense-in-depth security, routing AI shell execution into **network-disabled Docker containers** (`oven/bun:1`) with dropped Linux capabilities and **post-execution** `git diff` **snapshot auditing**.
- Integrated a **secondary AI verification agent** to automatically audit staged mutations for prompt injection defense, enforcing strict tool isolation policies and **10-minute task runtime locks**.

Snip.ly — URL Shortener — *Oct 2025*
React, Vite, Tailwind CSS, Cloudflare Workers, D1 (SQLite)

- Built a serverless URL redirection service on Cloudflare Workers (V8 isolates) and D1 (edge SQLite), delivering **sub-50ms redirects at the edge**.
- Hardened the API against SSRF with destination-host validation (blocking localhost, loopback, and RFC 1918 private IP ranges), origin-allowlisted CORS with `Vary: Origin`, and JWT verification via `jose` (Web Crypto).

Certifications

Oracle Cloud Infrastructure 2025 Generative AI Professional — Oracle — *Oct 2025*
Oracle Cloud Infrastructure 2025 Certified Developer Professional — Oracle — *Sep 2025*
Claude 101 — Anthropic — *Mar 2026*
Claude Code in Action — Anthropic — *Apr 2026*

Technical Skills

Languages: C/C++, TypeScript, JavaScript, Java, Python, SQL

Frameworks & Libraries: React, Node.js, Express.js, Bun, FastAPI, Tailwind CSS, TanStack Virtual, Zustand, Vercel AI SDK

Databases & Messaging: MySQL, PostgreSQL, SQLite (D1), MongoDB, Prisma ORM, BullMQ

Cloud, Security & DevOps: Docker, Linux, Cloudflare Workers, Playwright, Puppeteer, Git, GitHub